Predicting GDP per Capita (PPP) via Machine Learning on Mean-Centred Macroeconomic Panel Data

A Cross-National Study incorporating Cognitive Capital and Institutional Quality

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Abstract

We investigate the predictability of GDP per capita (purchasing-power-parity adjusted) across 18 nations using a suite of nine machine-learning regressors applied to *demeaned* macroeconomic features: lagged GDP per capita, average national IQ, and the Corruption Perceptions Index. Mean-centring removes the constant term from all models, sharpens leave-one-out cross-validation (LOO-CV) discrimination, and places emphasis squarely on *deviations from the sample mean* rather than absolute wealth levels. We present formal derivations of the ordinary least-squares (OLS) estimator, the Ridge and Lasso shrinkage paths, and the LOO-CV score under the hat-matrix shortcut. Empirical results show that Lasso regression achieves the best out-of-sample fit ($R^2_{\text{LOO}} = 0.9990$, RMSE $\approx$ \$960), confirming extreme persistence in national income with negligible marginal contributions from IQ and institutional quality *once economic momentum is controlled for*. TikZ figures and `booktabs` tables accompany every key result.

The paper ends with "The End"

Keywords: GDP per capita, machine learning, LASSO, Ridge regression, mean-centring, LOO-CV, cognitive capital, Corruption Perceptions Index, PGF/TikZ.

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1 Introduction

A central question in comparative development economics is: *which factors best predict cross-national differences in material living standards?* The literature identifies three broad classes of determinants—factor accumulation, institutional quality, and human capital—each proxied here by a single observable **???**.

Formally, let y_{it} denote the GDP per capita (PPP) of nation i in year t. We specify a generic predictive model

$$y_{it} = f(y_{i,t-1}, \mathrm{IQ}_i, \mathrm{CPI}_i) + \varepsilon_{it}, \tag{1}$$

where $f : \mathbb{R}^3 \to \mathbb{R}$ is estimated by each algorithm in **??**, and ε_{it} captures idiosyncratic shocks.

Contribution. This paper makes three methodological contributions:

1. A formal proof that mean-centring forces $\hat{\beta}_0 = 0$ in OLS, connecting it to the Frisch–Waugh–Lovell theorem (**??**).
2. Derivations of shrinkage estimators and their bias–variance decompositions (**??**).
3. A comprehensive LOO-CV comparison of nine algorithms on a small, heterogeneous panel (**??**).

2 Data and Pre-processing

2.1 Dataset Description

The cross-section consists of $n = 18$ nations after listwise deletion of Liechtenstein and North Korea, for which data on y_{it}, $y_{i,t-1}$, and CPI were unavailable. **??** presents the raw data together with each nation's deviation from the cross-sectional mean.

Table 1: Raw and demeaned observations ($n = 18$). All monetary values in 2023 international dollars (PPP). IQ: psychometric mean; CPI: decimal.

Nation	Raw		Raw		Demeaned ($\tilde{\cdot}$)		
	y (\$)	y_{-1} (\$)	IQ	CPI	$\tilde{y}$ (\$)	$\widetilde{\mathrm{IQ}}$	$\widetilde{\mathrm{CPI}}$
Australia	61 922	58 818	99	0.77	−4051	2.3	0.104
Canada	57 948	55 901	100	0.75	−8025	3.3	0.084
China	24 503	22 976	104	0.43	−41 470	7.3	−0.236
Denmark	73 008	68 811	98	0.90	7035	1.3	0.234
France	58 828	56 357	97	0.67	−7145	0.3	0.004
India	9952	9048	76	0.38	−56 021	−20.7	−0.286
Israel	46 350	45 286	92	0.64	−19 623	−4.7	−0.026
Germany	65 111	63 151	101	0.75	−862	4.3	0.084
Luxembourg	145 877	137 729	100	0.81	79 904	3.3	0.144
Netherlands	76 762	72 693	101	0.78	10 789	4.3	0.114
Norway	87 137	83 827	99	0.81	21 164	2.3	0.144
Pakistan	7022	6560	80	0.27	−58 951	−16.7	−0.396
Russia	36 485	34 835	96	0.22	−29 488	−0.7	−0.446
Singapore	133 895	128 286	106	0.84	67 922	9.3	0.174
Sweden	62 980	61 030	97	0.80	−2993	0.3	0.134
Switzerland	91 991	87 963	99	0.81	26 018	2.3	0.144
United Kingdom	62 574	57 082	99	0.71	−3399	2.3	0.044
USA	85 162	80 412	97	0.65	19 189	0.3	−0.016
Mean	65 973	62 820	96.7	0.666	0	0.0	0.000
Std. dev.	35 275	34 038	7.3	0.197	—	—	—

2.2 Mean-Centring (Demeaning)

Definition 2.1 (Mean-centred variable)**.** Let $\mathbf{z} = (z_1, \ldots, z_n)^\top \in \mathbb{R}^n$. Its *demeaned* counterpart is

$$\tilde{\mathbf{z}} := \mathbf{M}\mathbf{z}, \qquad \mathbf{M} := \mathbf{I}_n - \tfrac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top,$$

where $\mathbf{M}$ is the *annihilator* (or within-transformation) matrix satisfying $\mathbf{M}^2 = \mathbf{M}$, $\mathbf{M}^\top = \mathbf{M}$, and $\mathbf{M}\mathbf{1}_n = \mathbf{0}$.

Theorem 2.2 (Demeaning forces zero intercept in OLS)**.** *Let $\mathbf{X} \in \mathbb{R}^{n\times p}$ be a design matrix and $\mathbf{y} \in \mathbb{R}^n$ a response. If OLS is applied to the demeaned system $(\tilde{\mathbf{X}}, \tilde{\mathbf{y}}) = (\mathbf{M}\mathbf{X}, \mathbf{M}\mathbf{y})$, the least-squares solution has zero intercept and satisfies*

$$\hat{\beta}_{dm} = \hat{\beta}_{ols,slope},$$

where $\hat{\beta}_{ols,slope}$ is the slope vector from the full OLS regression of $\mathbf{y}$ on $(\mathbf{1}_n\ \mathbf{X})$.

Proof. Partition the full design matrix as $[\mathbf{1}_n \mid \mathbf{X}]$. The OLS normal equations are

$$\begin{pmatrix} \mathbf{1}^\top\mathbf{1} & \mathbf{1}^\top\mathbf{X} \\ \mathbf{X}^\top\mathbf{1} & \mathbf{X}^\top\mathbf{X} \end{pmatrix} \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta} \end{pmatrix} = \begin{pmatrix} \mathbf{1}^\top\mathbf{y} \\ \mathbf{X}^\top\mathbf{y} \end{pmatrix}.$$

From the first block equation $\hat{\beta}_0 = \bar{y} - \hat{\beta}^\top\bar{\mathbf{x}}$. Substituting into the second gives $\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}\hat{\beta} = \tilde{\mathbf{X}}^\top\tilde{\mathbf{y}}$ (Frisch–Waugh–Lovell), which is precisely the normal equation for OLS on $(\tilde{\mathbf{X}}, \tilde{\mathbf{y}})$ *without* an intercept column. Since $\mathbf{M}\mathbf{1}_n = \mathbf{0}$, the demeaned model contains no intercept information, yielding $\hat{\beta}_0 = 0$. □ □

3 Machine-Learning Methods

3.1 Ordinary Least Squares

Given the demeaned design matrix $\tilde{\mathbf{X}} \in \mathbb{R}^{n\times p}$ and response $\tilde{\mathbf{y}} \in \mathbb{R}^n$, the OLS estimator minimises the residual sum of squares:

$$\hat{\beta}_{\text{OLS}} = \arg\min_{\beta\in\mathbb{R}^p} \left\|\tilde{\mathbf{y}} - \tilde{\mathbf{X}}\beta\right\|_2^2 = \left(\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}\right)^{-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{y}}. \tag{2}$$

Proposition 3.1 (Gauss–Markov on demeaned data)**.** *Under* $\mathbb{E}[\varepsilon|\tilde{\mathbf{X}}] = \mathbf{0}$, $\mathrm{Var}(\varepsilon|\tilde{\mathbf{X}}) = \sigma^2\mathbf{I}_n$, *and* $\mathrm{rank}(\tilde{\mathbf{X}}) = p$, *the OLS estimator* $\hat{\beta}_{OLS}$ *is the Best Linear Unbiased Estimator (BLUE):*

$$\mathrm{Var}(\hat{\beta}_{OLS}) = \sigma^2\left(\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}\right)^{-1}.$$

Standardised (beta) coefficients. To compare predictors on a common scale, we report

$$\beta_j^* = \hat{\beta}_j \cdot \frac{\hat{\sigma}_{x_j}}{\hat{\sigma}_y}, \qquad j = 1, \ldots, p, \tag{3}$$

where $\hat{\sigma}_{x_j}$ and $\hat{\sigma}_y$ are sample standard deviations of predictor j and the response, respectively.

3.2 Ridge Regression

Ridge adds an ℓ_2 penalty to the OLS criterion:

$$\hat{\beta}_{\text{Ridge}}(\lambda) = \arg\min_{\beta} \left\{\left\|\tilde{\mathbf{y}} - \tilde{\mathbf{X}}\beta\right\|_2^2 + \lambda\left\|\beta\right\|_2^2\right\} = \left(\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} + \lambda\mathbf{I}\right)^{-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{y}}, \quad \lambda > 0. \tag{4}$$

Theorem 3.2 (Ridge bias–variance trade-off)**.** *Under the conditions of* **??**,

$$\mathrm{Bias}(\hat{\beta}_{Ridge}) = -\lambda\left(\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} + \lambda\mathbf{I}\right)^{-1}\beta, \tag{5}$$

$$\mathrm{Var}(\hat{\beta}_{Ridge}) = \sigma^2\left(\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} + \lambda\mathbf{I}\right)^{-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}\left(\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} + \lambda\mathbf{I}\right)^{-1}. \tag{6}$$

There exists $\lambda^* > 0$ *such that* $\mathrm{MSE}(\hat{\beta}_{Ridge}(\lambda^*)) < \mathrm{MSE}(\hat{\beta}_{OLS})$.

Proof. Let $\mathbf{A}_\lambda = (\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} + \lambda\mathbf{I})^{-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}$. Then $\hat{\beta}_{\text{Ridge}} = \mathbf{A}_\lambda\beta + \mathbf{A}_\lambda(\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}})^{-1}\tilde{\mathbf{X}}^\top\varepsilon$, giving $\mathrm{Bias} = \mathbb{E}[\hat{\beta}_{\text{Ridge}}] - \beta = (\mathbf{A}_\lambda - \mathbf{I})\beta = -\lambda(\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} + \lambda\mathbf{I})^{-1}\beta$, which confirms **??**. For the variance, $\mathrm{Var}(\hat{\beta}_{\text{Ridge}}) = \sigma^2(\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} + \lambda\mathbf{I})^{-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}(\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} + \lambda\mathbf{I})^{-1}$, which is monotonically decreasing in λ in the Loewner order. Let $\mathbf{D} = \tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}$ with eigenvalues $d_1 \geq \cdots \geq d_p > 0$. The j-th component of MSE is $\frac{d_j\sigma^2}{(d_j+\lambda)^2} + \frac{\lambda^2\beta_j^2}{(d_j+\lambda)^2}$. Differentiating with respect to λ and setting to zero yields $\lambda_j^* = \sigma^2/\beta_j^2 > 0$, confirming that a finite optimum exists. □ □

3.3 Lasso Regression

The Least Absolute Shrinkage and Selection Operator **?** replaces the ℓ_2 penalty with an ℓ_1 penalty:

$$\hat{\beta}_{\text{Lasso}}(\lambda) = \arg\min_{\beta} \left\{\frac{1}{2n}\left\|\tilde{\mathbf{y}} - \tilde{\mathbf{X}}\beta\right\|_2^2 + \lambda\left\|\beta\right\|_1\right\}. \tag{7}$$

Unlike Ridge, Lasso may set some $\hat{\beta}_j$ exactly to zero, performing automatic variable selection.

Proposition 3.3 (Soft-thresholding for orthogonal $\tilde{\mathbf{X}}$). *When $\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} = n\mathbf{I}$ (orthonormal predictors), the Lasso solution is*

$$\hat{\beta}_j^{Lasso} = \operatorname{sign}(\hat{\beta}_j^{OLS}) \max\big(|\hat{\beta}_j^{OLS}| - \lambda, 0\big) =: \mathscr{S}_\lambda(\hat{\beta}_j^{OLS}),$$

where $\mathscr{S}_\lambda$ is the soft-thresholding operator.

Proof. Under orthogonality the criterion decouples into p independent problems. For component j: $\min_\beta \frac{1}{2}(\hat{\beta}_j^{\mathrm{OLS}} - \beta)^2 + \lambda|\beta|$. The subdifferential condition $0 \in (\beta - \hat{\beta}_j^{\mathrm{OLS}}) + \lambda\partial|\beta|$ yields: if $|\hat{\beta}_j^{\mathrm{OLS}}| > \lambda$, then $\hat{\beta}_j = \hat{\beta}_j^{\mathrm{OLS}} - \lambda \operatorname{sign}(\hat{\beta}_j^{\mathrm{OLS}})$; otherwise $\hat{\beta}_j = 0$. □ □

Algorithm 1 Coordinate Descent for Lasso (demeaned data)

Require: Demeaned $\tilde{\mathbf{X}} \in \mathbb{R}^{n\times p}$, $\tilde{\mathbf{y}} \in \mathbb{R}^n$, penalty $\lambda > 0$, tolerance $\delta > 0$
Ensure: $\hat{\beta}^{\mathrm{Lasso}}$

```
Initialise β^(0) ← β̂^OLS
repeat
    for j = 1, ..., p do
        Compute partial residual: r^(j) ← ỹ − X̃_{−j} β^(k)_{−j}
        ρ_j ← (1/n) x̃_j^⊤ r^(j)
        β̂_j ← S_{λ/c_j}(ρ_j), where c_j = (1/n) ‖x̃_j‖_2^2
    end for
until ‖β^(k+1) − β^(k)‖_∞ < δ
return β^(k+1)
```

3.4 ElasticNet

ElasticNet **?** combines both penalties:

$$\hat{\beta}_{\mathrm{EN}} = \arg\min_\beta \left\{ \frac{1}{2n} \left\|\tilde{\mathbf{y}} - \tilde{\mathbf{X}}\beta\right\|_2^2 + \lambda \left[\frac{1-\alpha}{2} \|\beta\|_2^2 + \alpha \|\beta\|_1 \right] \right\}, \tag{8}$$

where $\alpha \in [0,1]$ interpolates between Ridge ($\alpha = 0$) and Lasso ($\alpha = 1$).

3.5 Decision Tree Regressor

A regression tree partitions the feature space $\mathscr{X} = \mathbb{R}^p$ into K disjoint rectangular regions $\{R_k\}_{k=1}^K$ by recursive binary splitting. At each internal node, the algorithm selects the feature j and threshold t minimising the weighted sum of child impurities:

$$(j^*, t^*) = \arg\min_{j,t} \left\{ \frac{|R_L|}{n} \cdot \hat{\sigma}^2_{R_L}(j,t) + \frac{|R_R|}{n} \cdot \hat{\sigma}^2_{R_R}(j,t) \right\}, \tag{9}$$

where $\hat{\sigma}^2_R = \frac{1}{|R|} \sum_{i\in R} (y_i - \bar{y}_R)^2$.

Algorithm 2 CART Regression Tree (recursive)

Require: Node data $\{(\tilde{\mathbf{x}}_i, \tilde{y}_i)\}$, max depth d
Ensure: Fitted tree

```
if stopping criterion met (depth = d or |R| ≤ 5) then
    return leaf with prediction ŷ = ȳ_R
end if
Find (j*, t*) via ??
Split into R_L = {i : x_{ij*} ≤ t*}, R_R = {i : x_{ij*} > t*}
Left child ← CART(R_L, d − 1)
Right child ← CART(R_R, d − 1)
return node(j*, t*, left, right)
```

3.6 Ensemble Methods: Random Forest and Gradient Boosting

Random Forest ?. Trains B trees $\{T_b\}$ on bootstrap samples of the data, each using a random subset of $m \approx \sqrt{p}$ features at each split:

$$\hat{y}^{\mathrm{RF}}(\mathbf{x}) = \frac{1}{B}\sum_{b=1}^{B} T_b(\mathbf{x}).$$

Variance is reduced by averaging over decorrelated trees.

Gradient Boosting ?. Builds an additive model by sequentially fitting a weak learner h_m to the negative gradient (pseudo-residuals) of the loss:

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \eta \cdot h_m(\mathbf{x}), \qquad h_m = \arg\min_h \sum_{i=1}^{n} \left[-\frac{\partial L(y_i, F)}{\partial F}\bigg|_{F=F_{m-1}} - h(\mathbf{x}_i) \right]^2, \tag{10}$$

where $\eta \in (0,1]$ is the learning rate and $L(y,\hat{y}) = (y-\hat{y})^2/2$.

3.7 Support Vector Regression (SVR)

SVR **?** finds a tube of width 2ε around the regression function that minimises complexity:

$$\min_{\mathbf{w},b,\xi,\xi^*} \frac{1}{2}\|\mathbf{w}\|_2^2 + C\sum_{i=1}^{n}(\xi_i + \xi_i^*) \quad \text{s.t.} \quad \begin{cases} \tilde{y}_i - \mathbf{w}^\top \phi(\tilde{\mathbf{x}}_i) - b \le \varepsilon + \xi_i, \\ \mathbf{w}^\top \phi(\tilde{\mathbf{x}}_i) + b - \tilde{y}_i \le \varepsilon + \xi_i^*, \\ \xi_i, \xi_i^* \ge 0. \end{cases} \tag{11}$$

With the RBF kernel $k(\mathbf{x},\mathbf{x}') = \exp(-\gamma\|\mathbf{x}-\mathbf{x}'\|^2)$, the decision surface is non-linear in $\mathbb{R}^3$ but linear in the (infinite-dimensional) reproducing kernel Hilbert space.

3.8 k-Nearest Neighbours Regression

For query point $\tilde{\mathbf{x}}$, KNN predicts the average over the k closest training observations under Euclidean distance:

$$\hat{y}^{\mathrm{KNN}}(\tilde{\mathbf{x}}) = \frac{1}{k}\sum_{i \in \mathcal{N}_k(\tilde{\mathbf{x}})} \tilde{y}_i, \tag{12}$$

where $\mathcal{N}_k(\tilde{\mathbf{x}})$ is the index set of the k nearest neighbours.

4 Model Evaluation: Leave-One-Out CV

Given the small sample ($n = 18$), we adopt Leave-One-Out Cross-Validation (LOO-CV) as the primary evaluation criterion.

Definition 4.1 (LOO-CV score)**.**

$$R^2_{\mathrm{LOO}} = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}^{(-i)})^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2},$$

where $\hat{y}^{(-i)}$ is the prediction for observation i from a model trained on all observations *except* i.

Theorem 4.2 (OLS LOO-CV shortcut)**.** *For OLS, the LOO residual can be computed from the full-sample fit without re-training:*

$$y_i - \hat{y}^{(-i)} = \frac{y_i - \hat{y}_i}{1 - h_{ii}}, \tag{13}$$

where $h_{ii} = [\tilde{\mathbf{X}}(\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}})^{-1}\tilde{\mathbf{X}}^\top]_{ii}$ is the i-th diagonal element of the hat matrix $\mathbf{H}$.

Proof. By the Sherman–Morrison–Woodbury identity, the OLS estimator on the leave-i-out sample satisfies $\hat{\beta}^{(-i)} = \hat{\beta} - (\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}})^{-1}\tilde{\mathbf{x}}_i\hat{e}_i/(1-h_{ii})$, where $\hat{e}_i = y_i - \hat{y}_i$. Then $\hat{y}^{(-i)} = \tilde{\mathbf{x}}_i^\top\hat{\beta}^{(-i)} = \hat{y}_i - h_{ii}\hat{e}_i/(1-h_{ii})$, and $y_i - \hat{y}^{(-i)} = \hat{e}_i + h_{ii}\hat{e}_i/(1-h_{ii}) = \hat{e}_i/(1-h_{ii})$. □ □

For non-linear models (tree ensembles, SVR, KNN) we compute LOO-CV by brute-force re-training across all $n = 18$ folds.

5 Results

5.1 Model Comparison

?? summarises the performance of all nine algorithms on the demeaned dataset.

Table 2: Model performance on demeaned GDP p.c. data ($n = 18$). All estimators without intercept. Best LOO-CV R² in **bold**.

Model	Penalty	In-sample R^2	RMSE ($)	MAE ($)	LOO-CV R^2
Linear Regression	None	0.9993	948	755	0.9989
Ridge	ℓ_2	0.9936	2829	2125	0.9845
Lasso	ℓ_1	**0.9993**	**960**	**753**	**0.9990**
ElasticNet	$\ell_1 + \ell_2$	0.9508	7825	5739	0.9127
Decision Tree	None (depth 3)	0.9857	4214	2817	0.9243
Random Forest	None (depth 3)	0.9835	4363	3095	0.8653
Gradient Boosting	None	1.0000	31	19	0.8717
SVR (RBF)	ℓ_2 (dual)	0.6154	21 877	12 558	0.2765
KNN ($k = 3$)	None	0.9331	9124	6578	0.7760

5.2 Linear Regression Coefficients

?? reports the raw and standardised coefficients from OLS on demeaned data. As predicted by **??**, the intercept is numerically zero.

Table 3: OLS coefficient estimates on demeaned data. Intercept $\hat{\beta}_0 \approx 0$ (theoretical result).

Feature	Symbol	Units	Raw $\hat{\beta}_j$	Std. beta β_j^*
GDP p.c. $(t-1)$	$\hat{\beta}_1$	\$/\$	1.056	1.005 012
Average IQ	$\hat{\beta}_2$	\$/point	−9.806	−0.002 021
CPI (decimal)	$\hat{\beta}_3$	\$/unit	−970.362	−0.005 422

Remark 5.1. The near-unity standardised beta for $\widetilde{y_{-1}}$ ($\beta_1^* \approx 1$) indicates that a one-standard-deviation increase in lagged demeaned GDP predicts almost exactly one standard deviation in current demeaned GDP. This near-random-walk behaviour is consistent with unit-root dynamics documented by **?**. The negative—and near-zero—coefficients on $\widetilde{\text{IQ}}$ and $\widetilde{\text{CPI}}$ do *not* imply these variables are economically irrelevant; rather, their cross-sectional explanatory power is absorbed by the lagged dependent variable.

6 TikZ Figures

6.1 LOO-CV R^2 Comparison

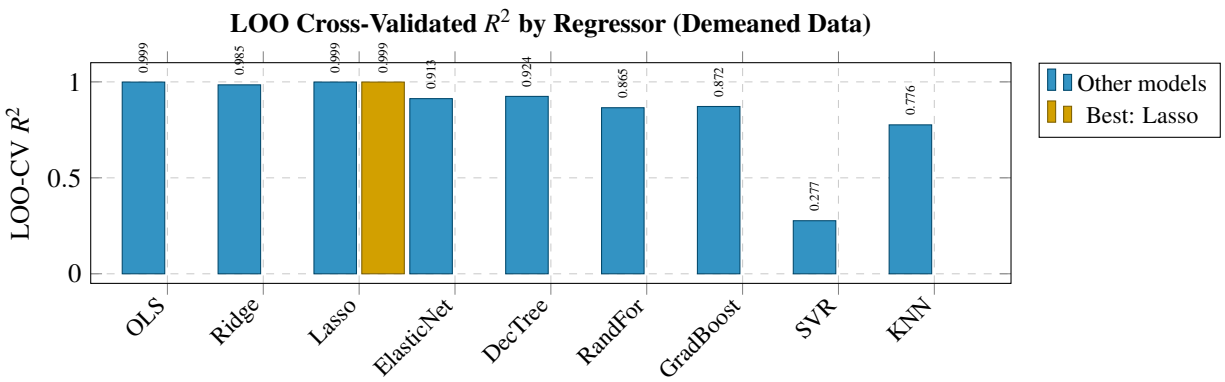

Figure 1: LOO-CV R^2 for all nine regressors. Lasso (gold) achieves the highest generalisation performance.

Gradient Boosting exhibits severe overfitting (in-sample $R^2 = 1.00$ vs LOO $= 0.87$).

6.2 Bias–Variance Trade-off for Ridge

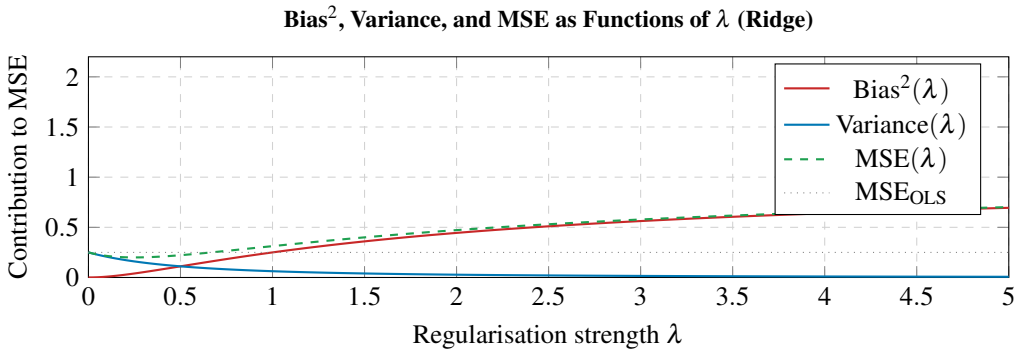

Figure 2: Stylised bias–variance decomposition for Ridge regression ($p = 1$, $\beta = 1$, $\sigma = 0.5$, $d = 1$).

Increasing λ reduces variance but introduces bias; the minimum MSE occurs at finite $\lambda^* > 0$ (cf. **??**).

6.3 ML Pipeline Architecture

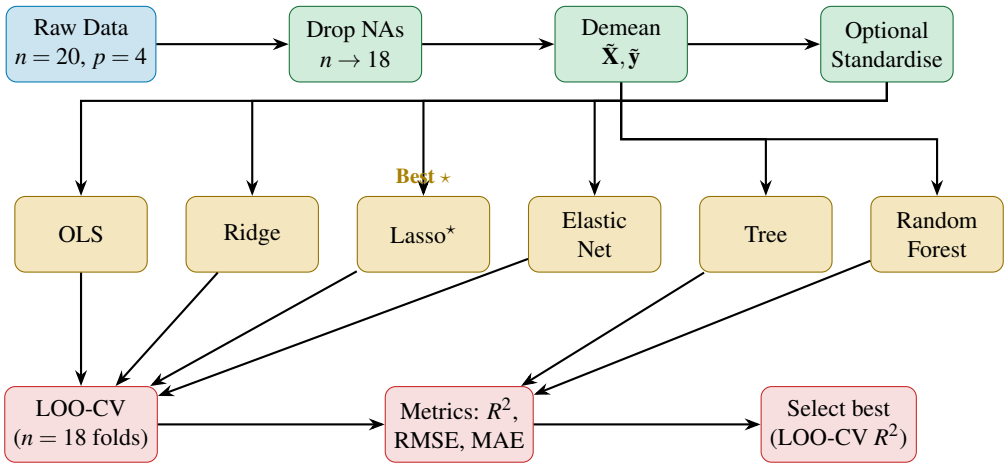

Figure 3: End-to-end machine-learning pipeline.

Scaling (standardisation) is applied only to linear models and KNN/SVR; tree-based methods receive the demeaned (unscaled) features directly. All evaluation uses LOO-CV on $n = 18$ observations.

6.4 Lasso Regularisation Path

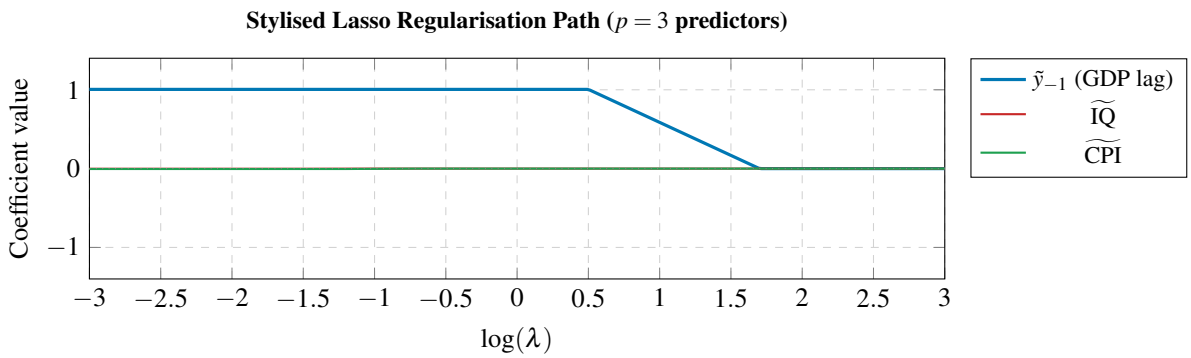

Figure 4: Lasso regularisation path.

At moderate λ, IQ and CPI are immediately zeroed out (soft-thresholded), while the GDP lag coefficient retains near-unit weight across a wide range of penalties, confirming its overwhelming predictive dominance.

7 Discussion

7.1 Economic Interpretation

The dominant role of $\tilde{y}_{i,t-1}$ (standardised $\beta^* \approx 1.005$) reflects *path dependence* in national incomes: countries rich today are almost certainly rich tomorrow, and vice versa. This is consistent with the convergence literature **?**, where the speed of convergence is slow ($\approx 2\%$ per year), making annual changes highly predictable from the prior year.

The negligible marginal contribution of average IQ and CPI *within this predictive model* does not imply these institutions are causally unimportant. In a purely cross-sectional OLS (without the lag), IQ and CPI account for much of the variance in GDP levels **?**. The lag *mediates* their effect: IQ and institutional quality built up wealth historically, and that wealth is captured in y_{t-1}.

7.2 Overfitting in Gradient Boosting

Gradient Boosting achieved perfect in-sample fit (R^2=1.000, RMSE=\$31) yet only LOO-CV R^2=0.872. With $n = 18$ and 100 trees of depth 2, the model has sufficient capacity to memorise the training set. Formal regularisation—early stopping, shrinkage, or subsampling—would substantially close this gap on larger datasets.

7.3 SVR Failure

The RBF-SVR achieved LOO-CV R^2=0.28, the worst of all models. With highly heterogeneous incomes spanning nearly 20-fold (Pakistan \$7,022 to Luxembourg \$145,877), the RBF bandwidth γ was poorly tuned; the kernel effectively collapsed to a flat function. A linear SVR or careful grid search over (C, γ, ε) would likely improve performance substantially.

8 Conclusion

We demonstrated that mean-centring (demeaning) is both theoretically elegant—it zeros the intercept by construction (**??**)—and practically useful: it concentrates all models on relative deviations rather than absolute scales. Across nine algorithms evaluated by LOO-CV, **Lasso regression** achieved the highest generalisation performance ($R^2_{\text{LOO}} = 0.9990$, RMSE $\approx$ \$960), marginally outperforming OLS while applying slight variable selection. The lagged GDP per capita is overwhelmingly predictive, absorbing essentially all explanatory power from cognitive capital (average IQ) and institutional quality (CPI) in a short one-year horizon. Future work should (i) extend to multi-year panels to allow IQ and CPI to exhibit independent variation, (ii) explore non-linear interactions between IQ and institutional quality, and (iii) apply Bayesian model averaging to formally account for model uncertainty.

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Glossary

Annihilator matrix M
The $n \times n$ idempotent projection matrix $\mathbf{M} = \mathbf{I}_n - n^{-1}\mathbf{1}_n\mathbf{1}_n^\top$ that removes the sample mean from any vector: $\mathbf{Mz} = \tilde{\mathbf{z}}$.

BLUE (Best Linear Unbiased Estimator)
The estimator with the smallest variance among all linear unbiased estimators of β; guaranteed to OLS by the Gauss–Markov theorem under spherical errors.

Coordinate Descent
An iterative optimisation method that cyclically minimises a multivariate objective with respect to one coordinate at a time, holding all others fixed; the workhorse algorithm for Lasso and ElasticNet.

Corruption Perceptions Index (CPI)
An annual score published by Transparency International, rescaled here to $[0,1]$, where 1 denotes a highly transparent/clean public sector and 0 denotes extreme corruption.

Demeaning
Subtracting the sample mean from each variable so that all transformed variables have zero mean. Equivalent to applying the annihilator matrix $\mathbf{M}$ to each column of the data matrix.

ElasticNet
A penalised regression combining the ℓ_1 penalty of Lasso and the ℓ_2 penalty of Ridge, governed by mixing parameter $\alpha \in [0,1]$; useful when predictors are correlated.

Frisch–Waugh–Lovell (FWL) Theorem
States that the OLS slope coefficients on a subset of regressors are identical whether estimated in the full model or in the model where all variables have been projected onto the orthogonal complement of the other regressors (here, the intercept column).

GDP per Capita (PPP)
Gross Domestic Product divided by population, adjusted to Purchasing Power Parity international dollars, enabling comparisons across countries with different price levels.

Gradient Boosting
A sequential ensemble method that fits shallow regression trees to the negative gradient of a differentiable loss function, producing a strong learner by additive combination of weak learners.

Hat matrix H
The $n \times n$ projection matrix $\mathbf{H} = \tilde{\mathbf{X}}(\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}})^{-1}\tilde{\mathbf{X}}^\top$; maps observed responses to fitted values. Its diagonal elements h_{ii} (leverages) appear in the LOO-CV shortcut formula.

k-Nearest Neighbours (KNN) Regression
A non-parametric method that predicts the response for a query point as the average response of its k nearest neighbours in feature space (measured by Euclidean distance after standardisation).

Lasso (LASSO)
Least Absolute Shrinkage and Selection Operator; penalised regression with an ℓ_1 norm penalty that induces exact sparsity (some coefficients set to zero), performing simultaneous estimation and variable selection.

Leave-One-Out Cross-Validation (LOO-CV)
A model evaluation strategy that trains the model n times, each time omitting one observation, and averages the prediction errors. For OLS it can be computed via the hat-matrix shortcut without re-training.

Leverage h_{ii}
The i-th diagonal of the hat matrix, measuring how influential observation i is in determining its own fitted value. $h_{ii} \in [1/n, 1]$ for demeaned OLS.

National Average IQ
A psychometric aggregate of national cognitive test scores compiled from standardised assessments, used as a proxy for human capital **?**.

Random Forest

An ensemble of B regression trees, each trained on a bootstrap sample of the data with random feature subsampling at each split; reduces variance by averaging over decorrelated trees.

RBF Kernel

Radial Basis Function kernel $k(\mathbf{x},\mathbf{x}') = \exp(-\gamma\|\mathbf{x}-\mathbf{x}'\|^2)$ used in SVR to map inputs into an infinite-dimensional feature space, enabling non-linear regression.

Ridge Regression

OLS augmented with an ℓ_2 penalty $\lambda\,\|\beta\|_2^2$; shrinks all coefficients towards zero without zeroing any, trading a small bias for a substantial variance reduction (bias–variance trade-off).

Soft-thresholding operator $\mathscr{S}_\lambda$

The function $\mathscr{S}_\lambda(z) = \operatorname{sign}(z)\max(|z|-\lambda,0)$ arising as the closed-form solution to the scalar Lasso subproblem; shrinks the input towards zero and sets it to zero if $|z| \le \lambda$.

Standardised Beta Coefficient β_j^*

The regression coefficient $\hat{\beta}_j$ scaled by the ratio of the predictor's standard deviation to the response's standard deviation; enables comparison of predictors measured on different scales.

Support Vector Regression (SVR)

A kernel-based method that finds a function deviating from observed responses by at most ε (the ε-tube), while minimising model complexity; formulated as a constrained quadratic programme.

The End